

School of Computer Science and Engineering (SCOPE)

Fall Semester 2025-26

CBS3005 - Cloud, Microservices and Applications

LAB ASSESSMENT 5

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22BBS0079

Task 1: Analyze API Activities using AWS CloudTrail

Description: Enable AWS CloudTrail to monitor and log all API activities in the AWS account. Also, analyze and detect unauthorized or unusual access.

Steps:

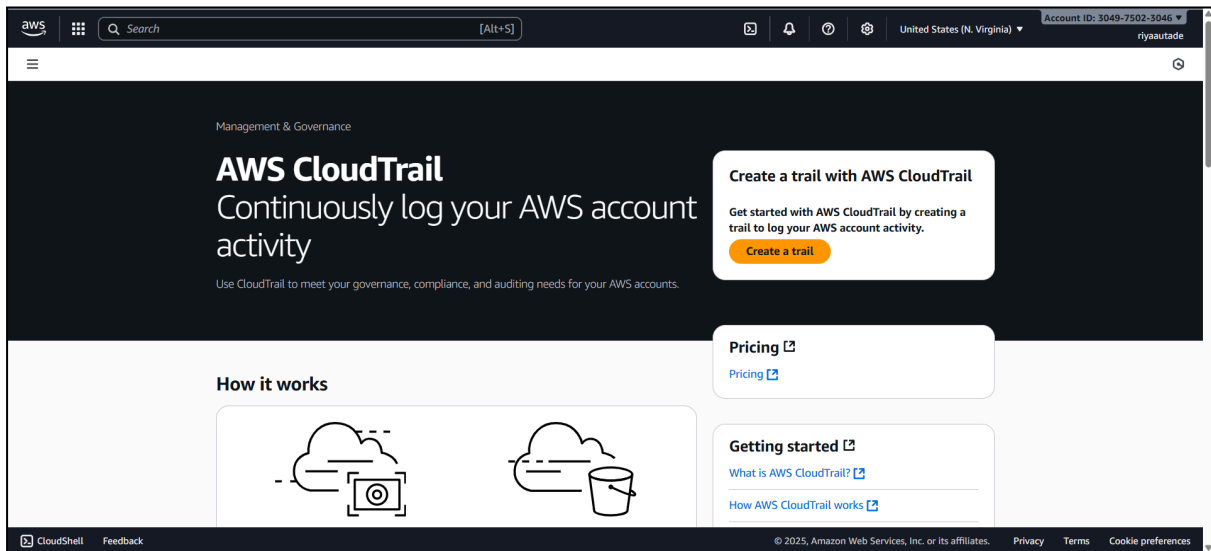
1. Go to CloudTrail Console → Trails → Create Trail.
2. Choose:
 - o Trail name: MySecurityTrail
 - o Apply trail to All regions
 - o Store logs in a new S3 bucket
 - o Enable CloudWatch Logs Integration (optional).
3. Perform a few activities in your AWS account:
 - o Launch or stop an EC2 instance.
 - o Create an S3 bucket.
4. Return to CloudTrail → Event History and view recent events.
5. Identify:
 - o Who performed the action (IAM user or role)
 - o Time of action
 - o Source IP address
 - o Affected resource

Expected Output:

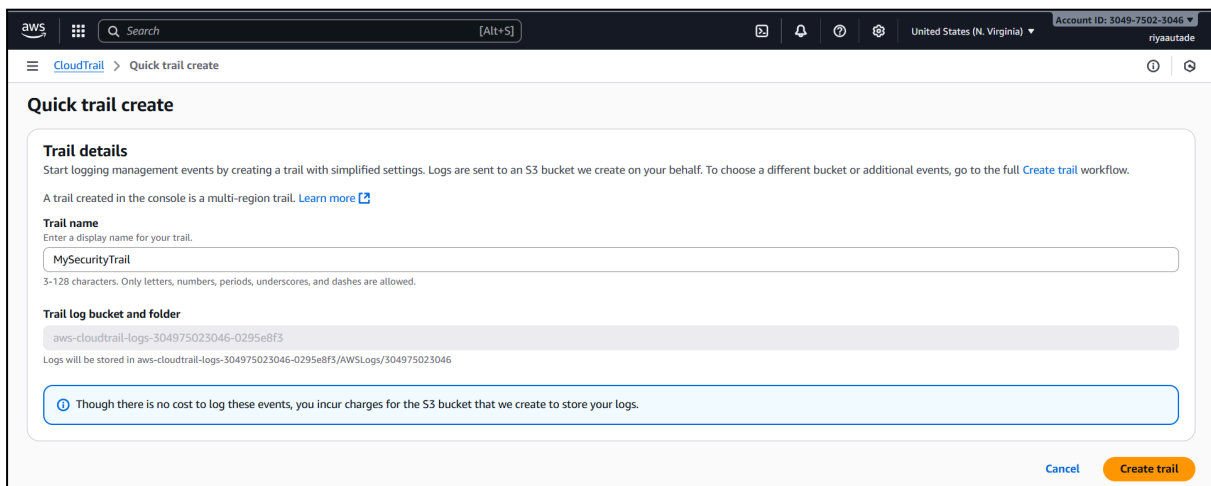
- Screenshot of CloudTrail event list.
- Sample JSON event record for one operation.
- Short explanation of how CloudTrail supports auditing and incident investigation

STEPS:

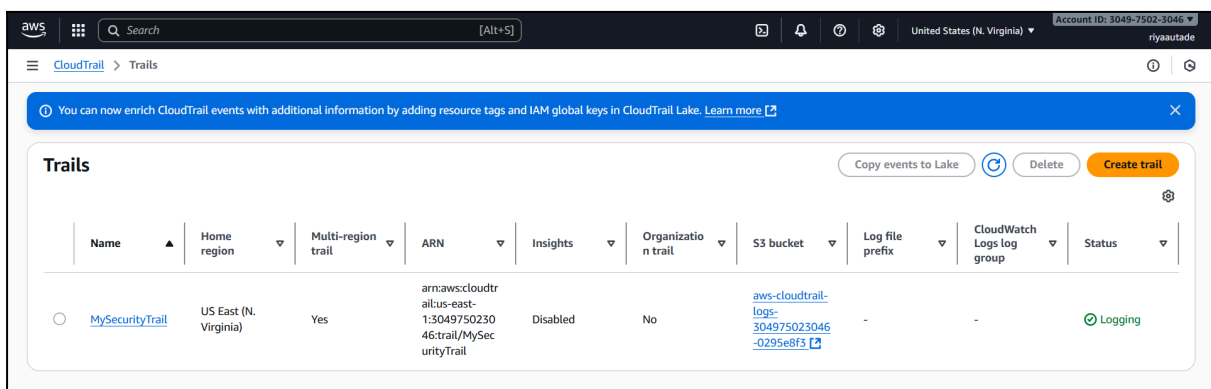
1. Go to CloudTrail and click on 'Create a trail'



2. Name your trail. Regions and S3 bucket is automatically handled. Click on create.



3. Trail is created. Click on it for details.



Trail in detail:

The screenshot shows the AWS CloudTrail console for a trail named 'MySecurityTrail'. The trail is currently logging. The console displays the following details:

- General details:**
 - Trail logging: Logging
 - Trail log location: [aws-cloudtrail-logs-304975023046-0295e8f3/AWSLogs/304975023046](#)
 - Log file validation: Disabled
 - SNS notification delivery: Disabled
 - Trail name: MySecurityTrail
 - Last log file delivered: -
 - Last file validation delivered: -
 - Multi-region trail: Yes
 - Log file SSE-KMS encryption: Not enabled
 - Apply trail to my organization: Not enabled
- CloudWatch Logs:** No CloudWatch Logs log groups. CloudWatch Logs is not configured for this trail.

Buttons for 'Delete', 'Stop logging', and 'Edit' are visible at the top right.

4. Create an EC2 instance to show activity

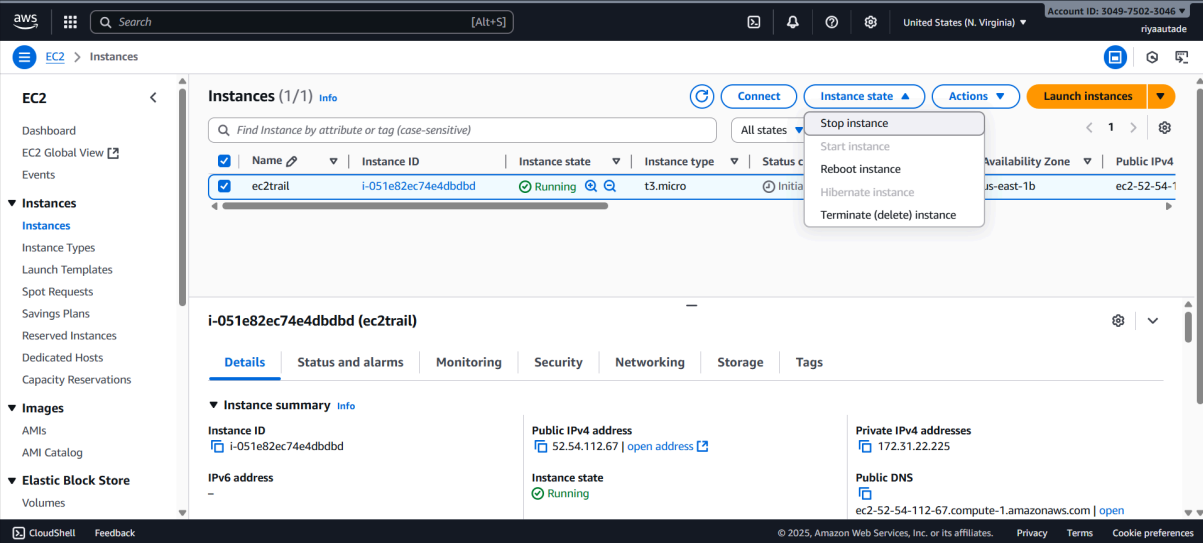
The screenshot shows the AWS Management Console for the Amazon Elastic Compute Cloud (EC2) service. The page includes the following sections:

- Header:** Amazon Elastic Compute Cloud (EC2) - Create, manage, and monitor virtual servers in the cloud.
- Launch a virtual server:** Buttons for 'Launch instance', 'View dashboard', 'Get started walkthroughs', and 'Get started tutorial'.
- Benefits and features:** EC2 offers ultimate scalability and control. Fully resizable compute capacity to support virtually any workload. This service is best if you want:
 - Highest level of control of the entire technology stack, allowing full integration with all AWS services
 - Widest variety of server size options
- Additional actions:** 'View running instances' and 'Migrate a server'.

The screenshot shows the AWS Management Console for the 'Launch an instance' page. A green success message indicates that the launch of instance `i-051e82ec74e4dbdbd` was successful. The page includes the following sections:

- Success:** Successfully initiated launch of instance `i-051e82ec74e4dbdbd`.
- Launch log:** A section for viewing the launch log.
- Next Steps:** A search bar for finding next steps, with a list of 6 items.
- Create billing usage alerts:** To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds. Button: 'Create billing alerts'.
- Connect to your instance:** Once your instance is running, log into it from your local computer. Button: 'Connect to instance'. Link: 'Learn more'.
- Connect an RDS database:** Configure the connection between an EC2 instance and a database to allow traffic flow between them. Button: 'Connect an RDS database'. Link: 'Create a new RDS database'. Link: 'Learn more'.
- Create EBS snapshot policy:** Create a policy that automates the creation, retention, and deletion of EBS snapshots. Button: 'Create EBS snapshot policy'.

5. Stop the EC2 instance



The screenshot shows the AWS Management Console for the 'Instances' page. The instance 'ec2trail' (ID: i-051e82ec74e4dbdbd) is in the 'Running' state. The 'Actions' dropdown menu is open, showing options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. The 'Stop instance' option is highlighted.

Name	Instance ID	Instance state	Instance type	Status check
ec2trail	i-051e82ec74e4dbdbd	Running	t3.micro	Initializing

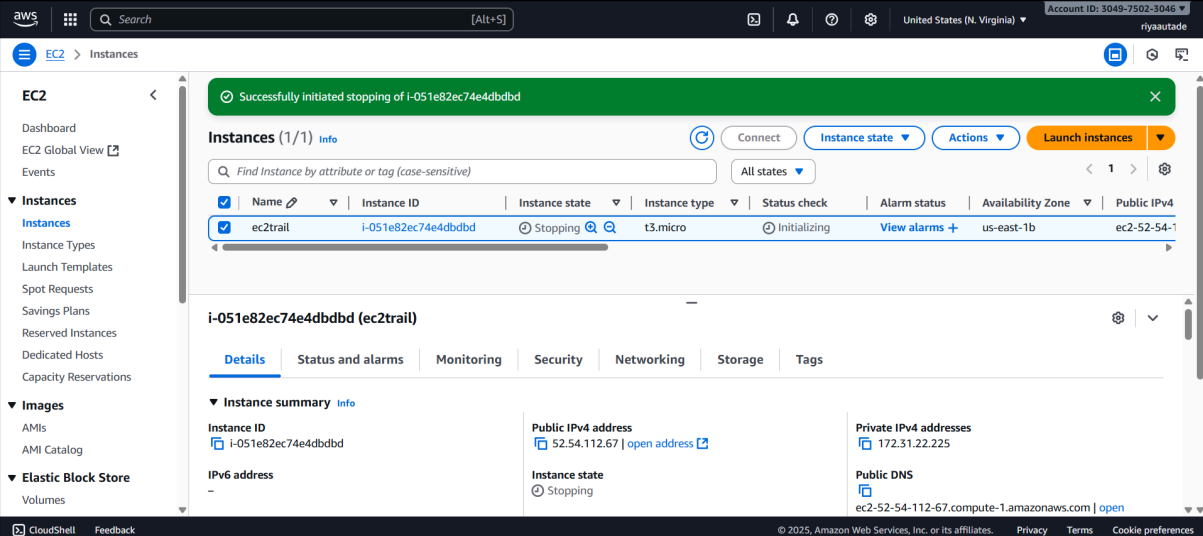
i-051e82ec74e4dbdbd (ec2trail)

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-051e82ec74e4dbdbd	52.54.112.67 open address	172.31.22.225

Instance state

Running



The screenshot shows the AWS Management Console for the 'Instances' page. A green banner at the top indicates 'Successfully initiated stopping of i-051e82ec74e4dbdbd'. The instance 'ec2trail' (ID: i-051e82ec74e4dbdbd) is in the 'Stopping' state. The 'Instance state' dropdown menu is open, showing options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. The 'Stop instance' option is highlighted.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
ec2trail	i-051e82ec74e4dbdbd	Stopping	t3.micro	Initializing	View alarms +	us-east-1b	ec2-52-54-1

i-051e82ec74e4dbdbd (ec2trail)

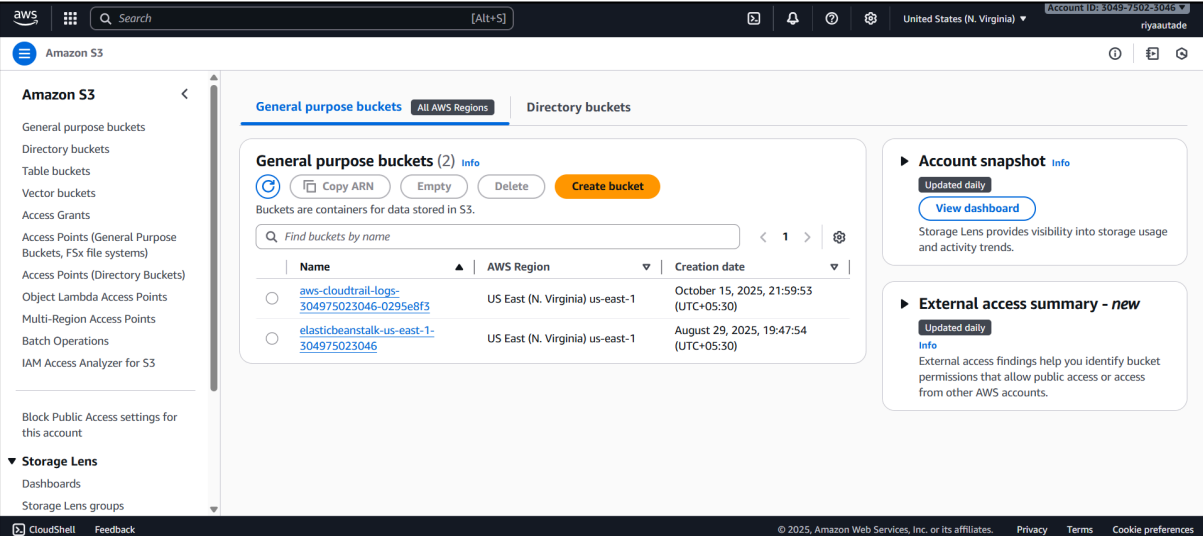
Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-051e82ec74e4dbdbd	52.54.112.67 open address	172.31.22.225

Instance state

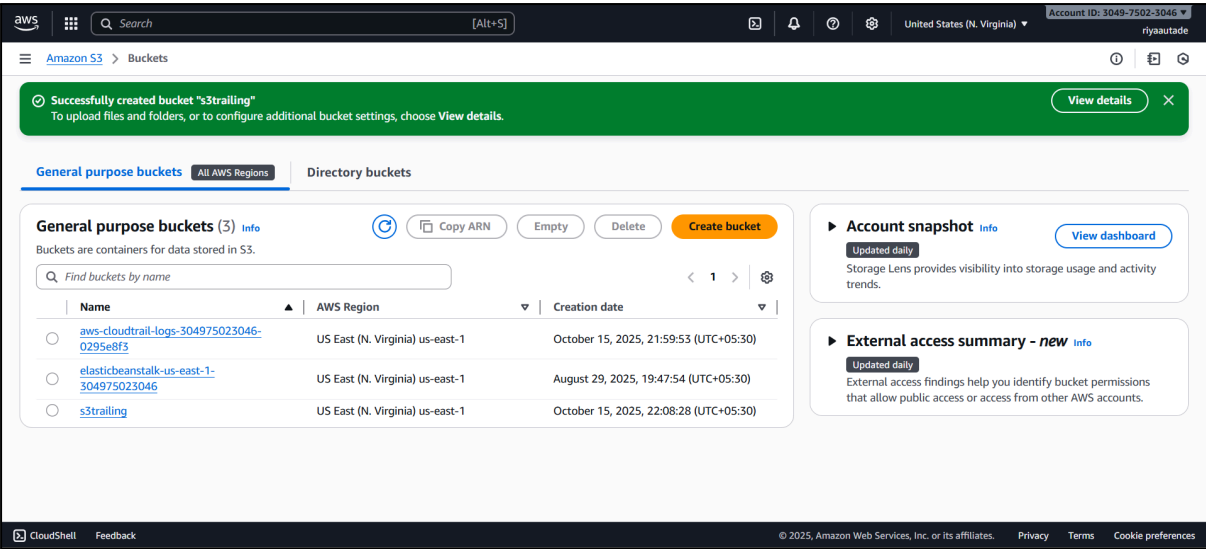
Stopping

6. Create an S3 bucket



The screenshot shows the AWS Management Console for the 'Amazon S3' page. The 'General purpose buckets' tab is selected. The 'Create bucket' button is highlighted. The table shows two existing buckets:

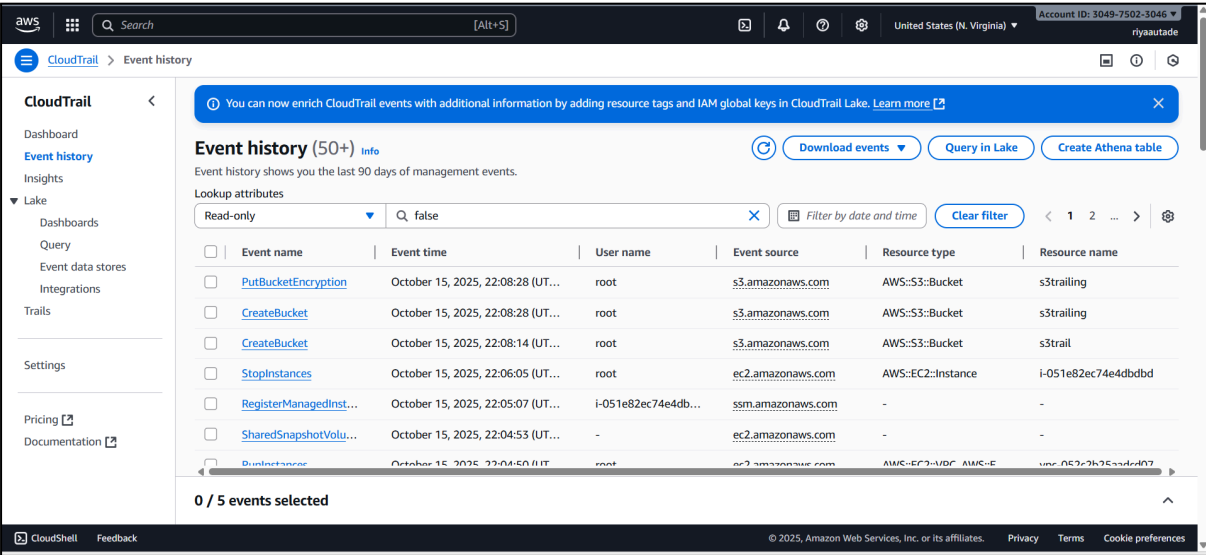
Name	AWS Region	Creation date
aws-cloudtrail-logs-304975023046-0295e8f3	US East (N. Virginia) us-east-1	October 15, 2025, 21:59:53 (UTC+05:30)
elasticbeanstalk-us-east-1-304975023046	US East (N. Virginia) us-east-1	August 29, 2025, 19:47:54 (UTC+05:30)



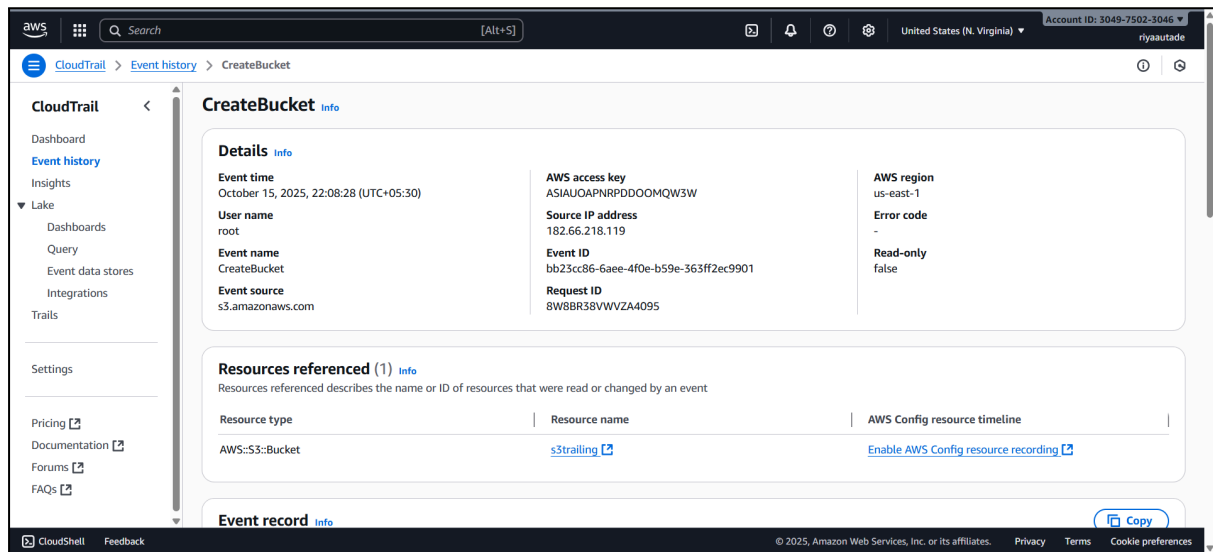
Go to CloudTrail and click on Event History to see all the recent activities including the EC2 instance creation, stopping and S3 bucket creation.

OUTPUT:

1. CloudTrail Event List:



For eg. CreateBucket event:



2. JSON Log for the event:

```
{
  "eventVersion": "1.11",
  "userIdentity": {
    "type": "Root",
    "principalId": "304975023046",
    "arn": "arn:aws:iam::304975023046:root",
    "accountId": "304975023046",
    "accessKeyId": "ASIAUOAPNRPDDOOMQW3W",
    "sessionContext": {
      "attributes": {
        "creationDate": "2025-10-15T16:24:30Z",
        "mfaAuthenticated": "true"
      }
    }
  },
  "eventTime": "2025-10-15T16:38:28Z",
  "eventSource": "s3.amazonaws.com",
  "eventName": "CreateBucket",
  "awsRegion": "us-east-1",
  "sourceIPAddress": "182.66.218.119",
  "userAgent": "[Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36]",
  "requestParameters": {
    "bucketName": "s3trailing",
    "Host": "s3trailing.s3.us-east-1.amazonaws.com"
  }
}
```

```

},
"responseElements": null,
"additionalEventData": {
  "SignatureVersion": "SigV4",
  "CipherSuite": "TLS_AES_128_GCM_SHA256",
  "bytesTransferredIn": 0,
  "AuthenticationMethod": "AuthHeader",
  "x-amz-id-2":
"G/HdNyS933puf0TO+9nw5jdiVqr2V7776Yyb0nuvwUv/2/7U/nBOP1E1ZhGeqAqBP0Ko2
/zknbZv0R4XrtV5nzA4KSX1YHounSd7BEiQjtU=",
  "bytesTransferredOut": 0
},
"requestID": "8W8BR38VWVZA4095",
"eventID": "bb23cc86-6aee-4f0e-b59e-363ff2ec9901",
"readOnly": false,
"eventType": "AwsApiCall",
"managementEvent": true,
"recipientAccountId": "304975023046",
"eventCategory": "Management",
"tlsDetails": {
  "tlsVersion": "TLSv1.3",
  "cipherSuite": "TLS_AES_128_GCM_SHA256",
  "clientProvidedHostHeader": "s3trailing.s3.us-east-1.amazonaws.com"
} }

```

3. Short Explanation:

How CloudTrail supports auditing and incident investigation:

AWS CloudTrail provides a complete log of all API calls made in your AWS account, whether through the console, SDKs, CLI, or other AWS services.

It records critical details such as who performed the action, when it occurred, from where (source IP), and which resources were affected.

This helps:

- **Auditing:** Verifying compliance and tracking resource changes over time.
 - **Security Monitoring:** Detecting unauthorized or unusual activities (e.g., access from unknown IPs).
 - **Incident Investigation:** Reconstructing actions leading up to a security event and identifying compromised credentials or misconfigurations.
-

Task 2: Configure CloudWatch for EC2 Instance Monitoring

Description: Create and monitor a simple EC2 instance using Amazon CloudWatch , configure alarms, monitor system metrics, and analyze performance data to understand how CloudWatch helps in proactive threat and performance monitoring.

Steps:

1. Launch a t2.micro EC2 instance (Free Tier eligible) running Amazon Linux 2.
2. Install and configure the CloudWatch Agent using:
`sudo yum install amazon-cloudwatch-agent -y`
`sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard`
3. Choose to monitor:
 - o CPU utilization
 - o Memory usage
 - o Disk I/O
4. Start the CloudWatch agent: `sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \ -a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s`
5. Go to CloudWatch Console → Metrics → EC2 and verify data collection.
6. Create a CloudWatch Alarm:
 - o Metric: CPUUtilization
 - o Condition: >70% for 2 consecutive periods
 - o Notification: via SNS topic (send email alert).

Expected Output:

- CloudWatch dashboard showing metrics (CPU, Memory).
- Alarm configuration screenshot.
- Email alert showing triggered event

STEPS:

Launch an EC2 instance

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The page is divided into several sections:

- Name and tags:** A text box contains 'ec2cloudwatch' and a link to 'Add additional tags'.
- Application and OS Images (Amazon Machine Image):** A search bar with the text 'Search our full catalog including 1000s of application and OS images'. Below it, there are tabs for 'Recents', 'My AMIs', and 'Quick Start'. The 'Quick Start' tab is active, showing a grid of AMIs including Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian.
- Summary:** A sidebar on the right containing configuration details:
 - Number of instances:** 1
 - Software Image (AMI):** Amazon Linux 2023 AMI 2023.9.2...read more
 - Virtual server type (instance type):** t3.micro
 - Firewall (security group):** default
 - Storage (volumes):** 1 volume(s) - 8 GiB
- Buttons:** 'Cancel' and 'Launch instance' (highlighted in orange) are at the bottom right. A 'Preview code' link is also present.

EC2 instance is running

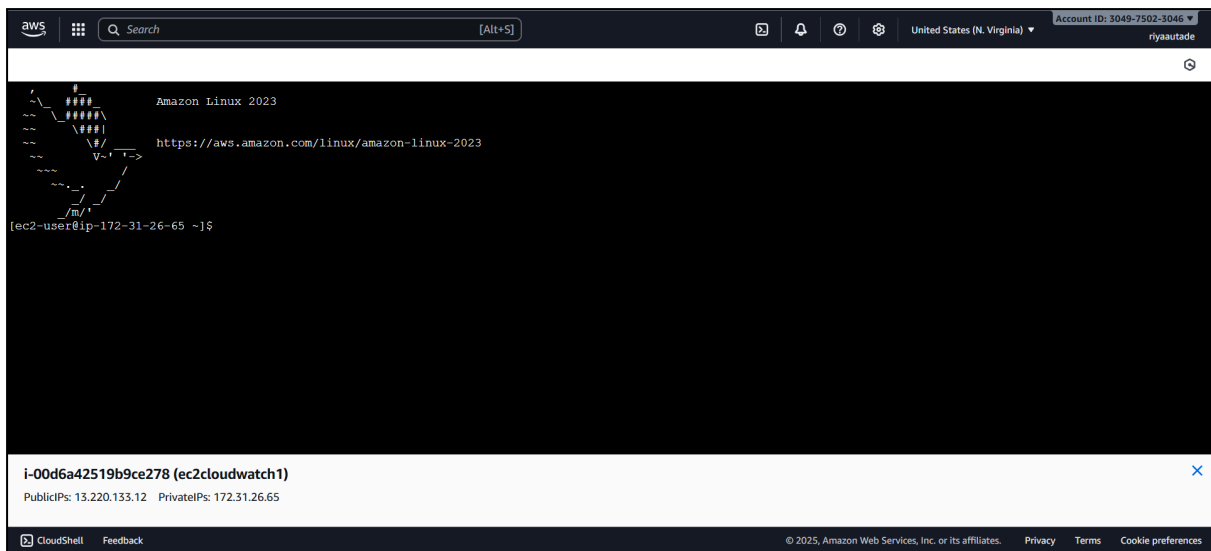
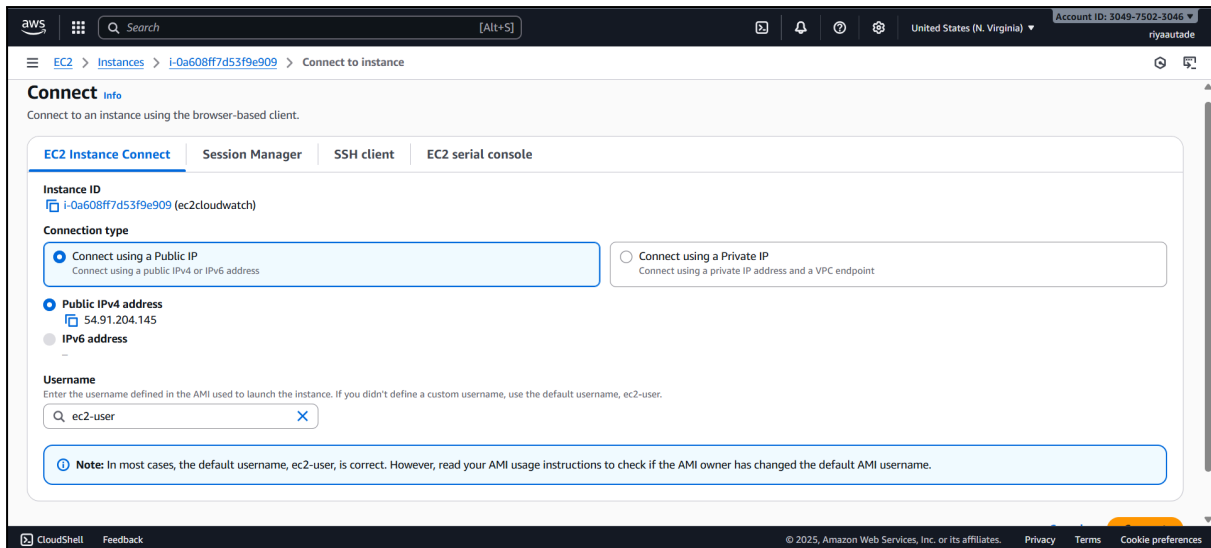
The screenshot shows the 'Launch an instance' page after successful completion. A green banner at the top reads 'Success Successfully initiated launch of instance (i-0a608ff7d53f9e909)'. Below this, there are several sections:

- Launch log:** A section for viewing the launch log.
- Next Steps:** A search bar with the text 'What would you like to do next with this instance, for example "create alarm" or "create backup"'. Below it, there are four cards with links to various actions:
 - Create billing usage alerts:** To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds. Link: 'Create billing alerts'.
 - Connect to your instance:** Once your instance is running, log into it from your local computer. Link: 'Connect to instance'. A 'Learn more' link is also present.
 - Connect an RDS database:** Configure the connection between an EC2 instance and a database to allow traffic flow between them. Link: 'Connect an RDS database'. A 'Create a new RDS database' link and a 'Learn more' link are also present.
 - Create EBS snapshot policy:** Create a policy that automates the creation, retention, and deletion of EBS snapshots. Link: 'Create EBS snapshot policy'.

2. Connect the EC2 instance (SSH)

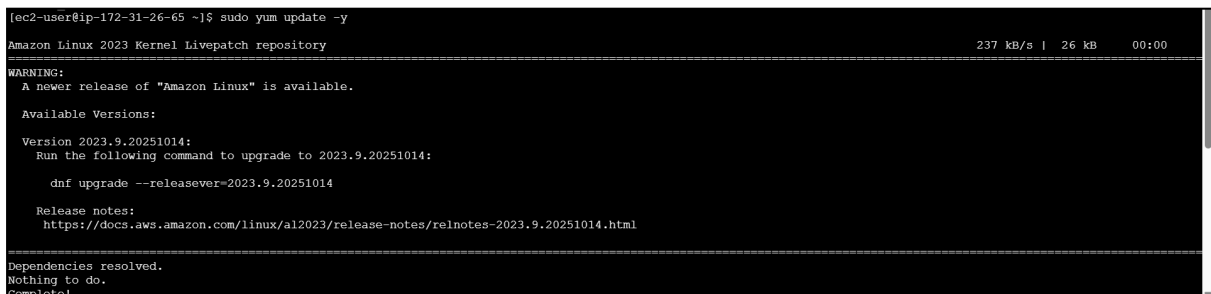
The screenshot shows the 'Instances' page in the AWS Management Console. The page displays a table of EC2 instances:

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	ec2trail	i-051e82ec74e4dbdbd	Stopped	t3.micro	-	View alarms +	us-east-1b	-
☒	ec2cloudwatch	i-0a608ff7d53f9e909	Running	t3.micro	Initializing	View alarms +	us-east-1b	ec2-54-91-1...



3. Install and configure CloudWatch Agent

`sudo yum update -y`



sudo yum install amazon-cloudwatch-agent -y

```
Running scriptlet: amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64
create group cwagent, result: 0
create user cwagent, result: 0

Installing      : amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64
Running scriptlet: amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64
Verifying      : amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64
=====
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.9.20251014:
    Run the following command to upgrade to 2023.9.20251014:

      dnf upgrade --releasever=2023.9.20251014

  Release notes:
    https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.9.20251014.html
=====

Installed:
  amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64

Complete!
```

sudo

/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard

```
[ec2-user@ip-172-31-26-65 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
=====
= Welcome to the Amazon CloudWatch Agent Configuration Manager =
=
= CloudWatch Agent allows you to collect metrics and logs from =
= your host and send them to CloudWatch. Additional CloudWatch =
= charges may apply.
=====
On which OS are you planning to use the agent?
1. linux
2. windows
3. darwin
default choice: [1]:
1
Trying to fetch the default region based on ec2 metadata...
I! imds retry client will retry 1 timesAre you using EC2 or On-Premises hosts?
1. EC2
2. On-Premises
default choice: [1]:
1
Which user are you planning to run the agent?
1. cwagent
2. root
3. others
default choice: [1]:
```

```

        "disk": {
            "measurement": [
                "used_percent"
            ],
            "metrics_collection_interval": 60,
            "resources": [
                "*"
            ]
        },
        "mem": {
            "measurement": [
                "mem_used_percent"
            ],
            "metrics_collection_interval": 60
        }
    }
}

```

Please check the above content of the config.
The config file is also located at /opt/aws/amazon-cloudwatch-agent/bin/config.json.
Edit it manually if needed.
Do you want to store the config in the SSM parameter store?
1. yes
2. no
default choice: [1]:
2
Program exits now.

4. Start the CloudWatch Agent

`sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \`
`-a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s`

```

[ec2-user@ip-172-31-26-65 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s
***** processing amazon-cloudwatch-agent *****
I! Trying to detect region from ec2 D! [EC2] Found active network interface I! imds retry client will retry 1 timesSuccessfully fetched the config and saved in /opt/aws/
amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_config.json.tmp
Start configuration validation...
2025/10/15 17:49:00 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_config.json.tmp ...
2025/10/15 17:49:00 I! Valid Json input schema.
2025/10/15 17:49:00 Configuration validation first phase succeeded
I! Detecting run as user...
I! Trying to detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -schematest -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml
Configuration validation second phase succeeded
Configuration validation succeeded
amazon-cloudwatch-agent has already been stopped
Created symlink /etc/systemd/system/multi-user.target.wants/amazon-cloudwatch-agent.service → /etc/systemd/system/amazon-cloudwatch-agent.service.

```

`sudo systemctl status amazon-cloudwatch-agent`

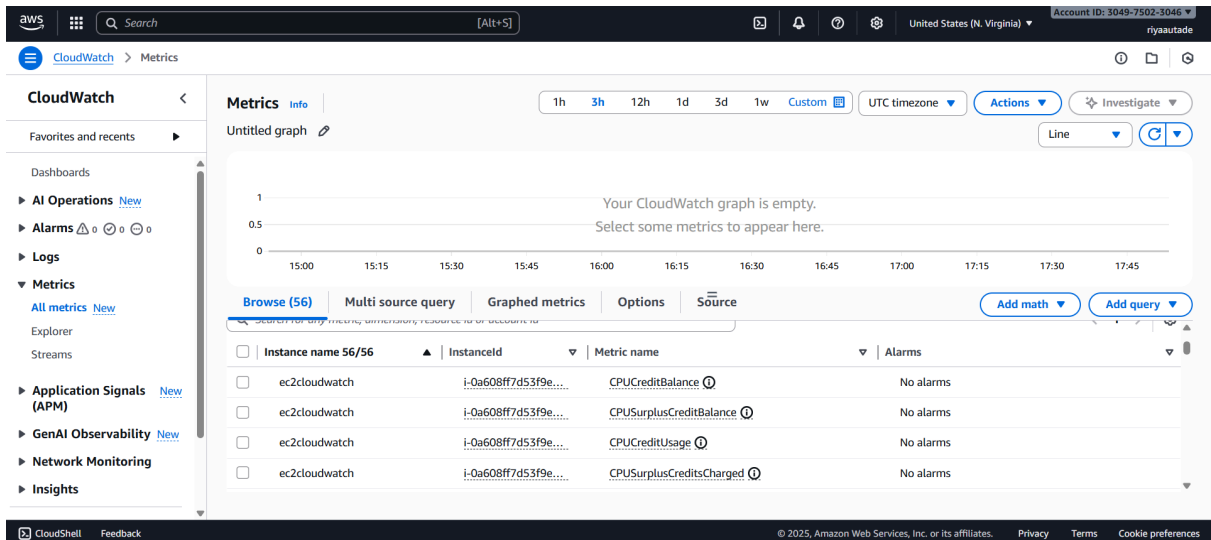
```

[ec2-user@ip-172-31-26-65 ~]$ sudo systemctl status amazon-cloudwatch-agent
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-10-15 17:49:02 UTC; 32s ago
     Main PID: 4040 (amazon-cloudwat)
        Tasks: 7 (limit: 1053)
      Memory: 49.0M
           CPU: 36ms
     CGroup: /system.slice/amazon-cloudwatch-agent.service
             └─4040 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml -envconfig /c

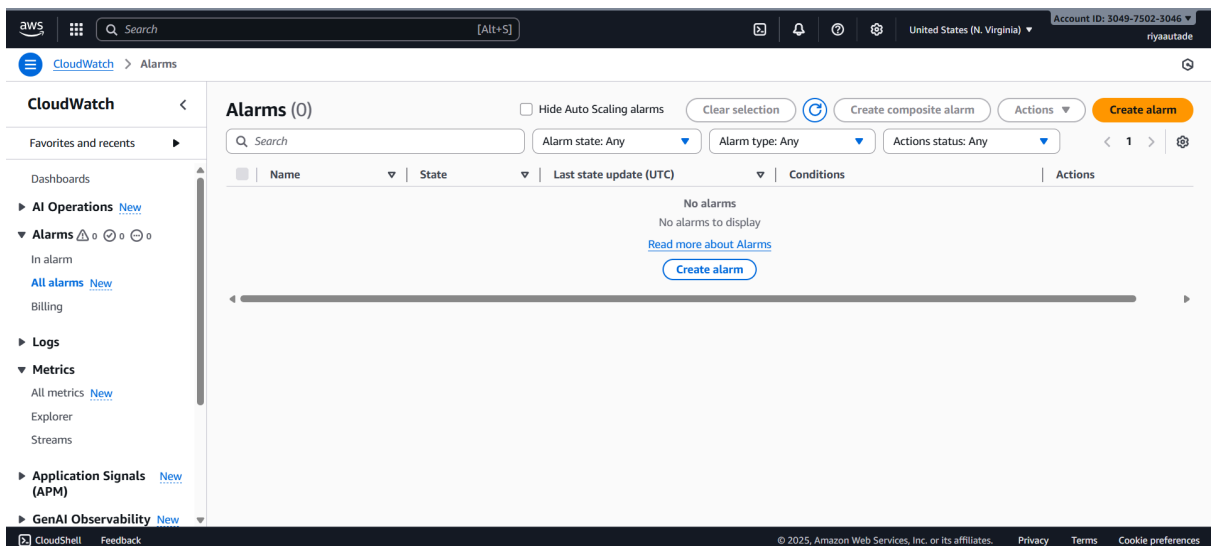
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: D! [EC2] Found active network interface
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: I! imds retry client will retry 1 timesI! Detected the instance is EC2
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: 2025/10/15 17:49:02 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json does not exist or ca
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: 2025/10/15 17:49:02 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: 2025/10/15 17:49:02 I! Valid Json input schema.
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: I! Detecting run as user...
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: I! Trying to detect region from ec2
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4045]: 2025/10/15 17:49:02 Configuration validation first phase succeeded
Oct 15 17:49:02 ip-172-31-26-65.ec2.internal start-amazon-cloudwatch-agent[4040]: I! Detecting run as user...
lines 1-20/20 (END)

```

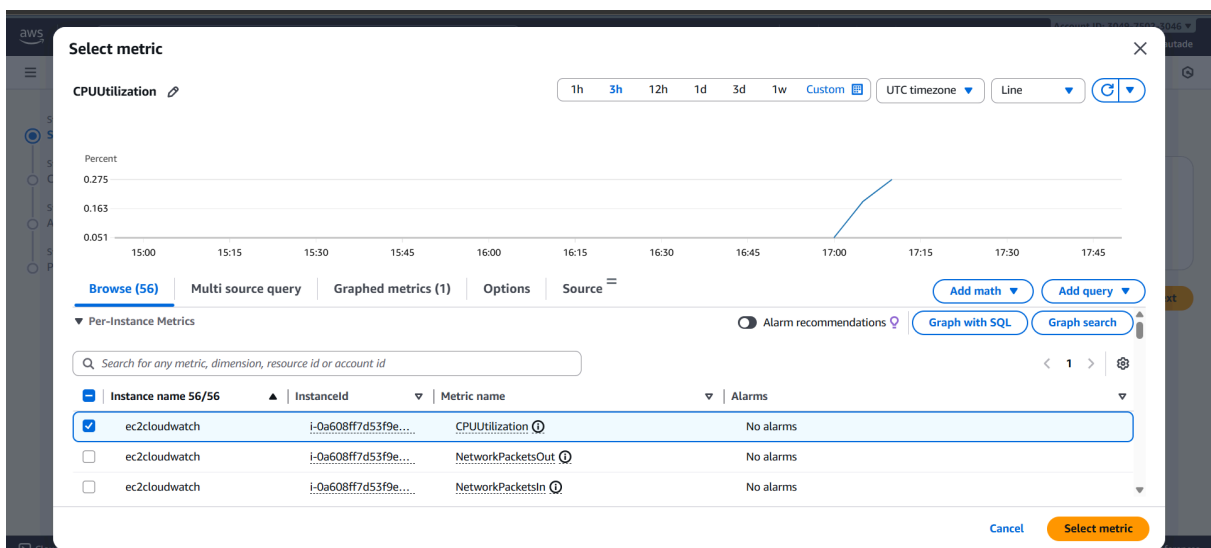
5. Go to CloudWatch Console → Metrics → EC2 and verify data collection.



6. Create alarms



Select CPU Utilization:



aws Search [Alt+S] United States (N. Virginia) Account ID: 3049-7502-3046 riyautade

CloudWatch > Alarms > Create alarm

Specify metric and conditions

Step 1: Specify metric and conditions
Step 2: Configure actions
Step 3: Add alarm details
Step 4: Preview and create

Metric

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.

Percent

0.275
0.163
0.051

15:00 15:30 16:00 16:30 17:00 17:30

CPUUtilization

Namespace
AWS/EC2

Metric name
CPUUtilization

InstanceId
i-0a608ff7d53f9e909

Instance name
ec2cloudwatch

Statistic
Average

Period
1 minute

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Threshold

aws Search [Alt+S] United States (N. Virginia) Account ID: 3049-7502-3046 riyautade

CloudWatch > Alarms > Create alarm

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...

Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...

Define the threshold value.

70

Must be a number.

Additional configuration

Cancel Next

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Create SNS topic and add email

aws Search [Alt+S] United States (N. Virginia) Account ID: 3049-7502-3046 riyautade

CloudWatch > Alarms > Create alarm

Step 3: Add alarm details
Step 4: Preview and create

Alarm state trigger

Define the alarm state that will trigger this action.

☒ In alarm
The metric or expression is outside of the defined threshold.

☐ OK
The metric or expression is within the defined threshold.

☐ Insufficient data
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic

Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ Create new topic

☐ Use topic ARN to notify other accounts

Create a new topic...

The topic name must be unique.

CPU_Utilization_DA

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...

Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

autaderiya@gmail.com

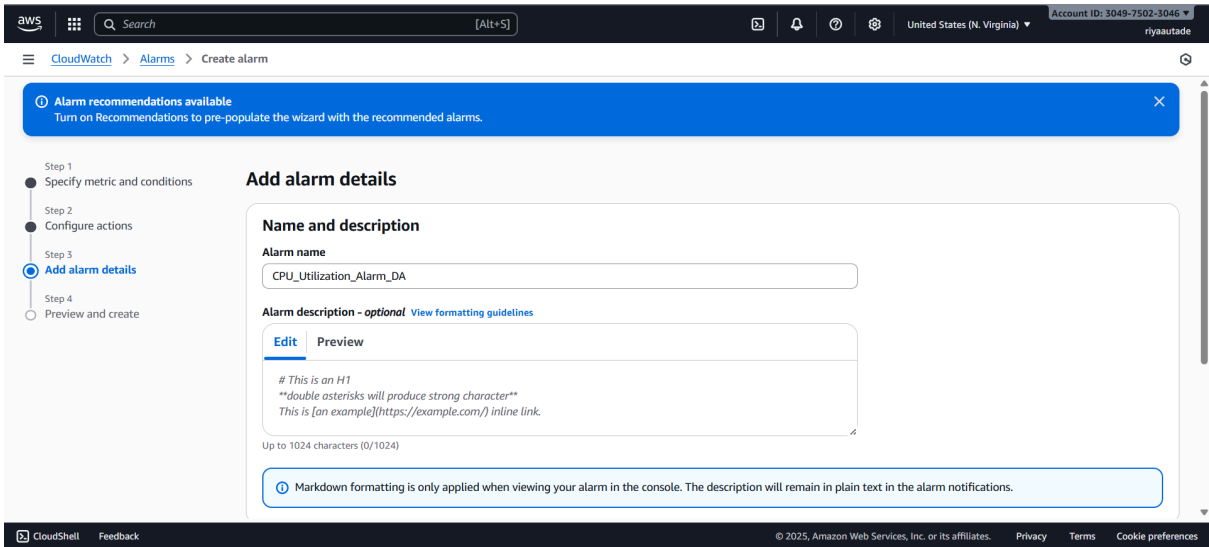
user1@example.com, user2@example.com

Create topic

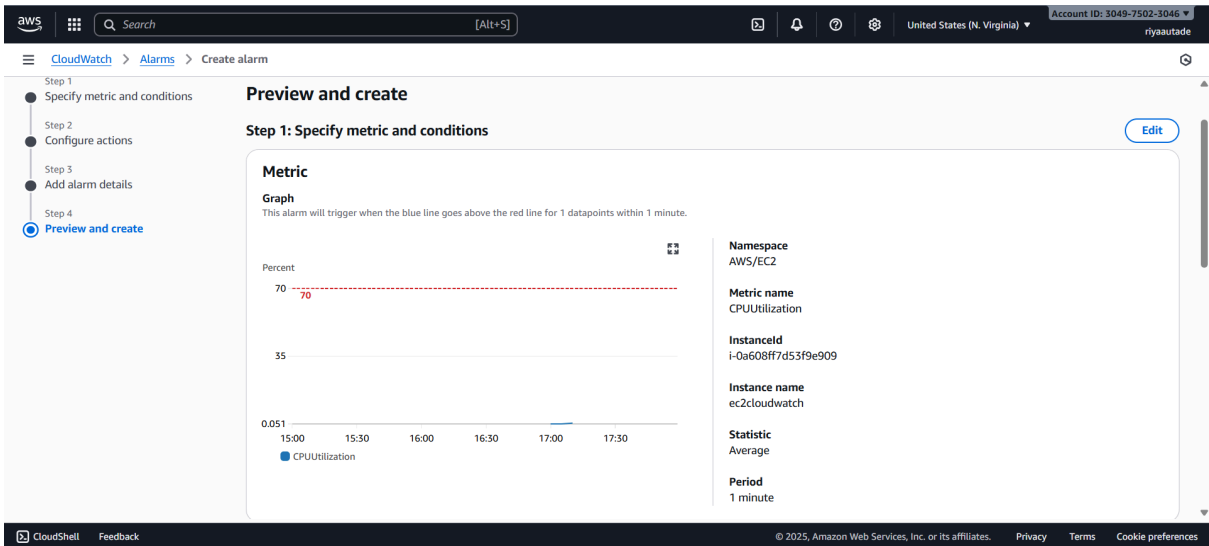
Add notification

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Name the Alarm



Preview and Create

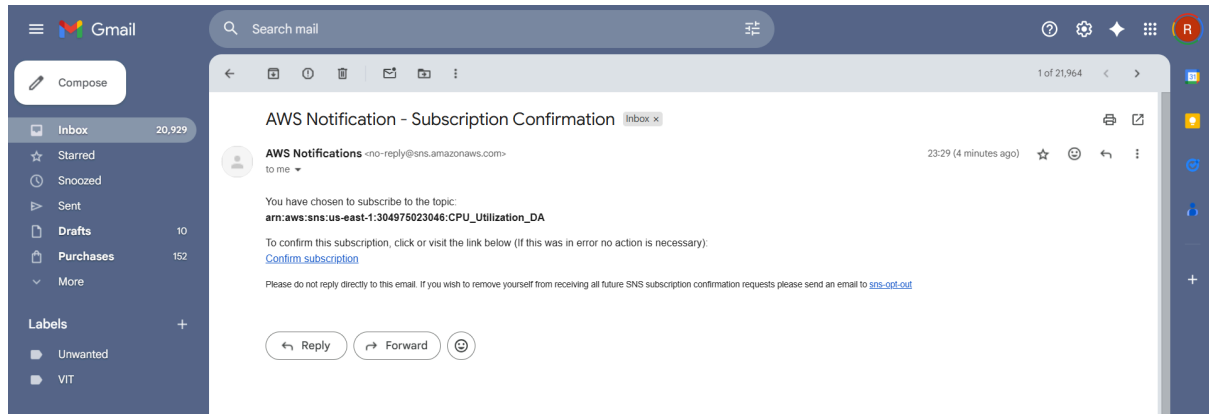


In Amazon SNS, confirm subscription

This screenshot shows the Amazon SNS 'Subscriptions' page. The left sidebar shows the 'Amazon SNS' navigation menu with 'Subscriptions' selected. The main area is titled 'Subscriptions (1)' and contains a table with one subscription. The table has columns for 'ID', 'Endpoint', 'Status', 'Protocol', and 'Topic'. The subscription is in 'Pending confirmation' status, with the endpoint 'autaderiya@gmail.com' and the topic 'CPU_Utilization_DA'. The bottom of the screen shows the AWS footer with '© 2025, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

ID	Endpoint	Status	Protocol	Topic
Pending confirmation	autaderiya@gmail.com	Pending confirmation	EMAIL	CPU_Utilization_DA

In email:



Simple Notification Service

Subscription confirmed!

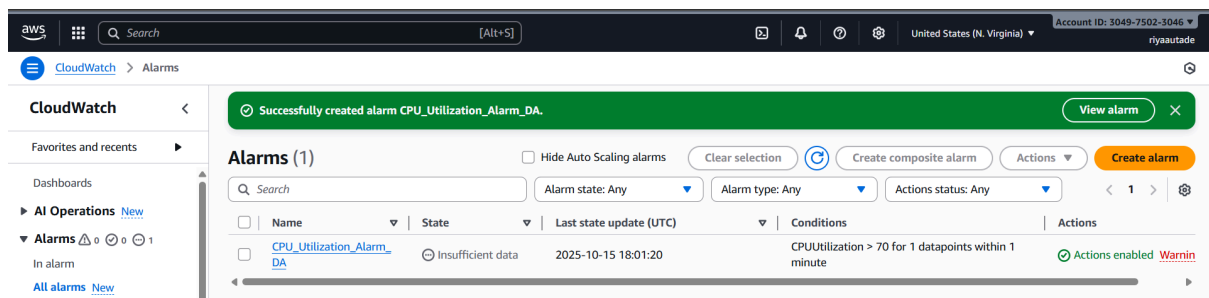
You have successfully subscribed.

Your subscription's id is:

arn:aws:sns:us-east-1:304975023046:CPU_Utilization_DA:01163333-1d14-4402-bc67-abf88b3b8d49

If it was not your intention to subscribe, [click here to unsubscribe](#).

Alarm created successfully:



7. Testing the Alarm: Spike CPU

Run this command to create high CPU load: `yes > /dev/null &`

Explanation:

yes outputs y continuously.

> /dev/null discards the output so it doesn't fill the terminal.

& runs the process in the background.

This will increase CPU utilization on your EC2 instance

Run command: top

```
top - 18:09:47 up 57 min, 3 users, load average: 1.13, 0.32, 0.11
Tasks: 110 total, 3 running, 107 sleeping, 0 stopped, 0 zombie
%Cpu(s): 59.0 us, 41.0 sy, 0.0 ni, 0.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 904.8 total, 61.7 free, 223.1 used, 620.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 551.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
5011	ec2-user	20	0	221372	1008	916	R	99.7	0.1	0:36.00	yes
5006	ec2-user	20	0	221372	1004	912	R	99.3	0.1	1:08.25	yes
4040	root	20	0	1430588	127508	81296	S	0.7	13.8	0:04.77	amazon-cloudwat
5013	ec2-user	20	0	224024	3380	2848	R	0.3	0.4	0:00.03	top
1	root	20	0	172656	17608	10812	S	0.0	1.9	0:01.76	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	slub_flushwq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	netns
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
11	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
14	root	20	0	0	0	0	S	0.0	0.0	0:00.07	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.08	rcu_preempt
16	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
18	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1

For CPU utilization

